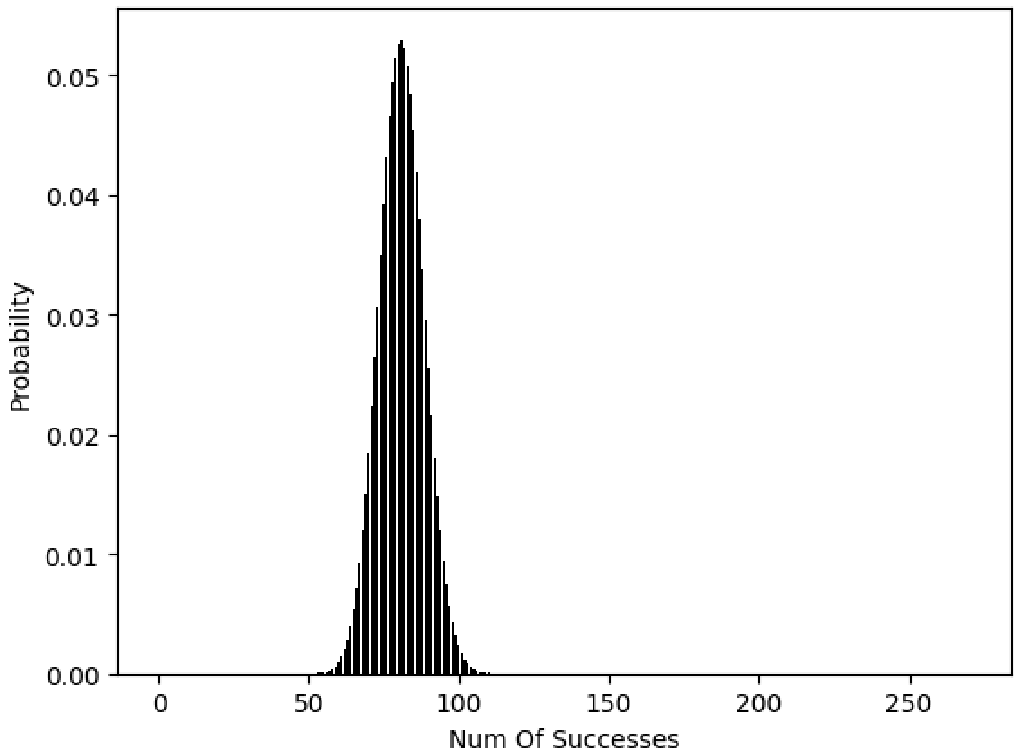
Question 1: Applying CLT On Bernoulli Distribution

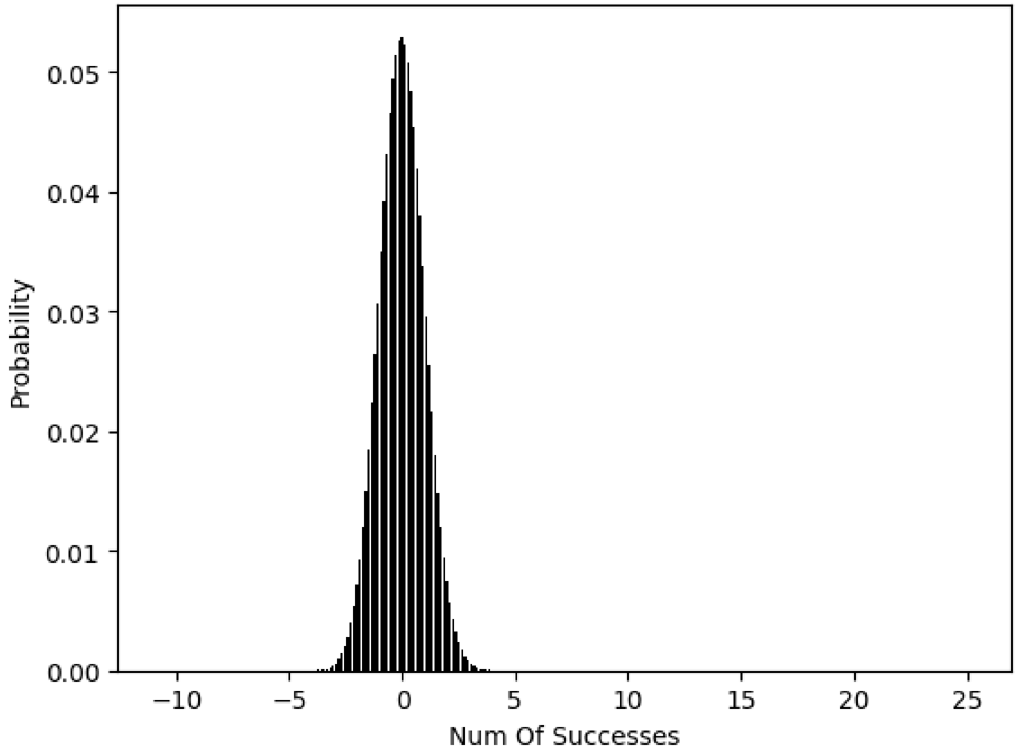
Part 1: Plotting The Initial Binomial Distribution

In this part I'll plot the pmf for X ~ Bin(270, 0.3) which is a binomial distribution. I'll use the np.scatter() function because the Binomial distribution is discrete and not continious to use plt.plot() function.



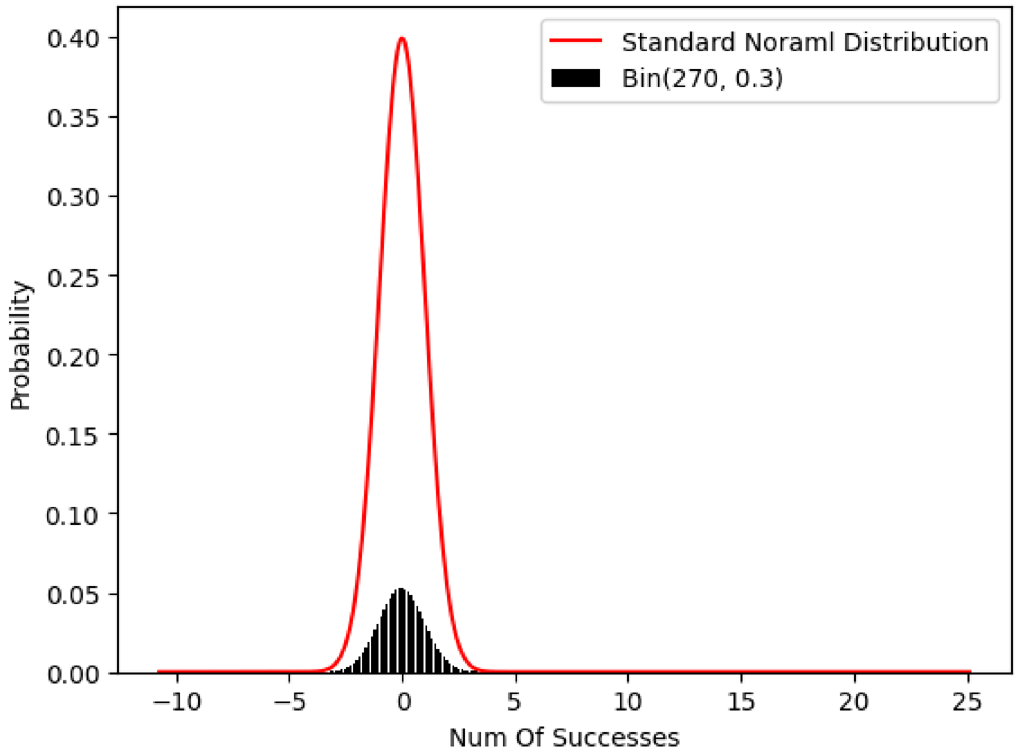
Part 2: Normalizing The X Axis

For this part we use the z-score normalization formula and normalize the x interval.



Part 3: Adding The Standard Normal Distribution To The Plot

I'll use the scipy library to to use its already built pdf function for normal distribution.



As we can see, these two distributions are so similar and it looks like they are only different in a coefficient. This gives us some intuition about the CLT theorom. Because the Binomial distribution is sum of some Bernoulli distributed variables.

Part 4: Sum Of Bars

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1.00000
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As we can see, the summation of the bar values in the pmf function for Bin(270, 0.3) is equal to 1. It is because the X random variable has to be in range of 0 to 270 and nothing else. But the noraml distribution diagram shows something else; it is a pdf function. It shows the probability density for our random variable not the probability itself. So actuall we can say each bar is an integral of that normal pdf in an interval of length 1. This justifies why the summation of all bars is equal to 1.

Part 5: Calculating And Imposing The Coefficient

To calculate the coefficient of normal pdf to estimate the Binomial distribuion, we have to be able to take the integral of that function from -5 to +5 and reverse it. But we can't because that function is not integrable. So we use the Riemann calculations.

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As we expected, the sum of values for riemann integral is equal to: 1.00
0.13280317881493256
0.13280317881493262
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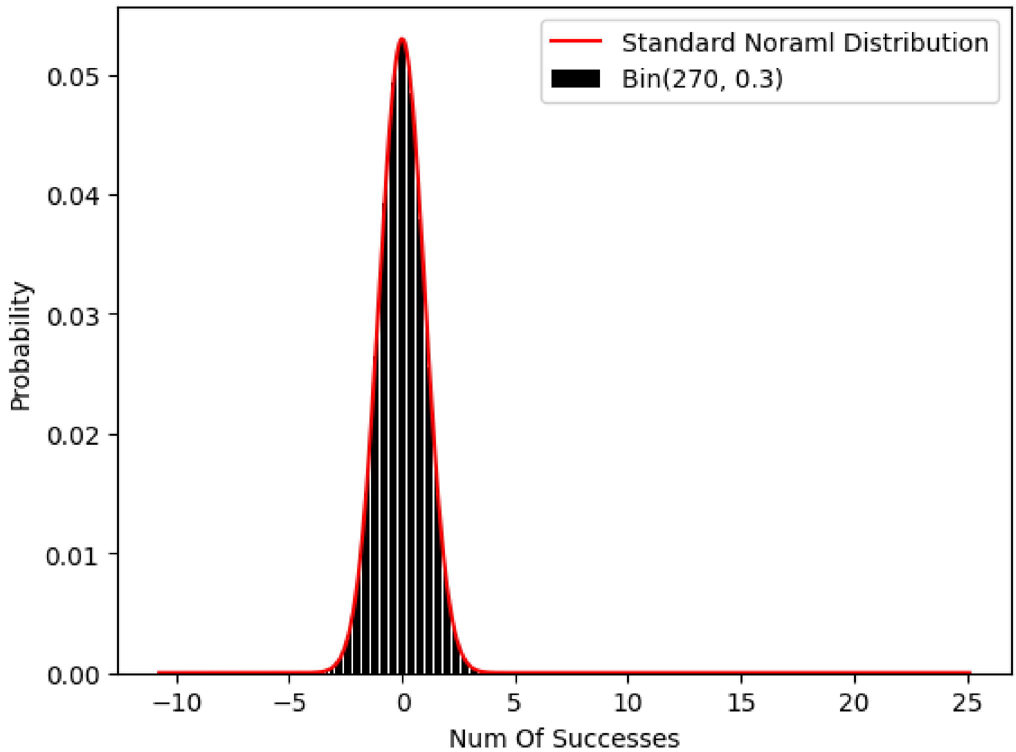
As we can see, balancing coefficient(denoted by C) is exactly equal to 1 / x_std. It means our coefficient is equal to reversed of standard deviation of the initial Binomial distributed random variable.



And here it is! A completely fit and satisfying estimation for the Binomial distribution.

Part 6: Coin Flipping: Binomial And Normal Estimation

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Binomial Probability:  0.048474296626430755
Normal Estimated Probability:  0.04839414490382868
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As we can see, the probabilities are exactly equal and they only have a slight and ignorable difference.

Part 7: Probability Of k Being Between 40 and 60

40 < k < 60 -> -2 < (k-50)/5 < 2

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0.9544994481518967
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